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School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 04-03-2026

ShopEase – Responsive E-Commerce Platform with State-Persistent Cart

SDG: SDG 8 – Decent Work and Economic Growth

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INTRODUCTION

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- ShopEase is a full-stack e-commerce web application built using Java Spring Boot and MySQL.
- It provides a responsive shopping interface where customers can browse products, manage a cart, and place orders.
- The cart is state-persistent — items saved in the cart remain even after the browser is closed or the device changes.
- Products are organized into categories like Electronics, Fashion, Books, Sports, and Home & Living.
- Users can register, login securely (BCrypt hashing), search/filter products, and track their order history.
- The backend exposes RESTful APIs (AuthServlet, ProductServlet, CartServlet, OrderServlet) consumed by the frontend.
- Two roles are supported: CUSTOMER (browse, buy) and ADMIN (manage products and categories).
- This project supports SDG 8 – Decent Work and Economic Growth by enabling small businesses to sell online efficiently.

PROBLEM STATEMENT

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- Many small businesses lack an affordable, easy-to-use online platform to sell their products.
- Existing e-commerce solutions are complex, expensive, or require technical expertise to set up.
- Customers face poor experiences like cart items disappearing after logout or browser close.
- There is no easy way to search, filter, and compare products across multiple categories in one place.
- Manual order processing leads to errors, delays, and customer dissatisfaction.
- Without secure authentication, user data and transactions are vulnerable to misuse.
- ShopEase addresses all these issues by providing a lightweight, secure, and user-friendly web platform.
- It enables users to shop online with a persistent cart, verified login, and seamless order placement.

OBJECTIVE(S)

Objective 1:

- To design and develop a responsive e-commerce web application (ShopEase) using Java Spring Boot and MySQL that enables customers to browse, search, and purchase products online.

Objective 2:

- To implement a secure, state-persistent shopping cart system backed by a relational database so that cart data is preserved across browser sessions and devices.

Objective 3:

- To provide role-based access control with CUSTOMER and ADMIN roles, secure BCrypt password authentication, and a RESTful API architecture for all application features.

PROPOSED SYSTEM

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- ShopEase is a Java-based e-commerce platform that allows users to register, login, and shop for products across multiple categories.
- The proposed system features a state-persistent cart stored in MySQL — cart items survive logout and browser restart.
- Products include detailed information: name, category, price, discount percentage, stock count, rating, and review count.
- Users can filter products by category, search by keyword, and sort by price or rating.
- The checkout process captures shipping address and payment method (COD), and places a confirmed order with stock deduction.
- BCrypt is used for password hashing, ensuring user passwords are never stored in plain text.
- Role-based system: CUSTOMER role accesses shopping features; ADMIN role manages the catalog and orders.
- The system uses RESTful APIs, making it easy to extend with mobile apps or third-party integrations in the future.

PROPOSED SYSTEM ARCHITECTURE

- **Architecture Overview: 3-Tier Web Application**
- Tier 1 - Presentation (Frontend): HTML, CSS, JavaScript pages served as static files from Spring Boot. Users interact via browser on port 8080.
- Tier 2 - Business Logic (Backend): Java 17 + Spring Boot 3.2.5 with Jakarta Servlets. Four controllers handle API routes: /api/auth, /api/products, /api/cart, /api/orders.
- Tier 3 - Data Layer: MySQL 8.x database (shopease) with 6 tables: users, categories, products, cart, orders, order_items.
- **Data Flow:**
- Browser sends HTTP requests -> Spring Boot routes to matching Servlet -> Servlet calls DAO -> DAO queries MySQL -> JSON response returned to frontend.
- **Security Layer:**
- ApiFilter validates session before allowing cart/order operations. BCrypt verifies password on login. Sessions expire after 30 minutes of inactivity.
- DBConnection utility class manages MySQL JDBC connections with URL: jdbc:mysql://localhost:3306/shopease

WORKFLOW

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- ShopEase User Workflow – Step-by-Step Data Flow:

- **Step 1: User Registration / Login**
- New user fills registration form -> POST /api/auth/register -> UserDAO stores BCrypt-hashed password in MySQL users table. Existing user logs in via POST /api/auth/login -> session created with userId, userName, userRole.
- **Step 2: Browse Products**
- Browser calls GET /api/products with optional category, search, and sort parameters. ProductDAO queries MySQL with dynamic SQL and returns JSON list of products.
- **Step 3: Add to Cart**
- User clicks Add to Cart -> POST /api/cart with productId. CartServlet checks session (must be logged in) -> CartDAO inserts or increments quantity in cart table using ON DUPLICATE KEY UPDATE.
- **Step 4: Checkout & Order Placement**
- User submits shipping address -> POST /api/orders. OrderServlet fetches cart items, calculates total, starts DB transaction: inserts order, inserts order_items, reduces stock, clears cart -> commits. Returns orderId.
- **Step 5: View Order History**
- User visits orders page -> GET /api/orders -> OrderDAO retrieves all orders with nested order_items for the logged-in user. Session verified by ApiFilter before processing.

TECHNOLOGIES USED

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Frontend:

- HTML5, CSS3, JavaScript (Vanilla) — served as static files from Spring Boot; responsive layout using Flexbox/Grid.

Backend:

Java 17, Spring Boot 3.2.5 with embedded Apache Tomcat; Jakarta Servlet API for REST endpoint controllers.

Database:

- MySQL 8.x — tables: users, categories, products, cart, orders, order_items. InnoDB with foreign key constraints.

Libraries & Tools:

- Google Gson 2.10.1 (JSON serialization), jBCrypt 0.4 (password hashing), MySQL Connector/J 8.2.0.

Build Tool:

- Apache Maven 3.x with Spring Boot Maven Plugin — packaged as an executable JAR file.

Development Tools:

- Eclipse IDE with Spring Tools, VS Code; Postman for API testing; MySQL Workbench for database management.

MODULE DESCRIPTION

ShopEase has 4 main modules as described below:

Module 1: User Authentication & Profile Management

- Handles registration (email uniqueness check), login with BCrypt, session management (30-min), and logout. Supports CUSTOMER and ADMIN roles via /api/auth/* endpoints.

Module 2: Product & Category Management

- Lists all products with filters (category, keyword, sort). Returns featured products and product details. Managed via /api/products/* endpoints backed by ProductDAO and MySQL.

Module 3: State-Persistent Shopping Cart

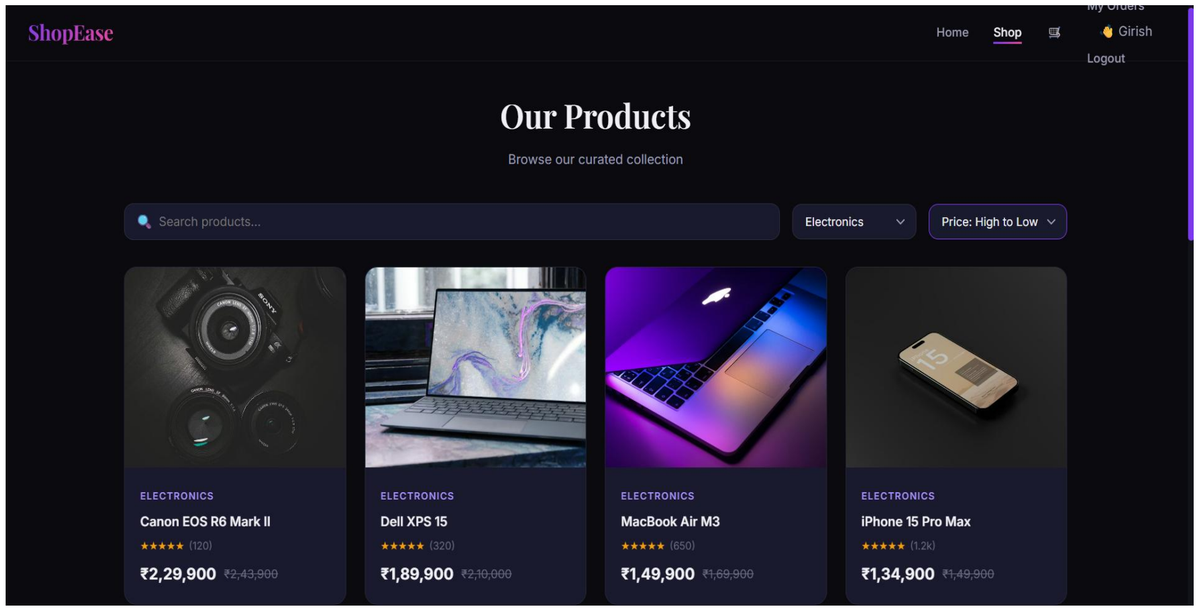
- Stores cart items in MySQL linked to user ID. Uses ON DUPLICATE KEY UPDATE for add. Supports update, remove, and clear operations. Cart survives logout via /api/cart/* endpoints.

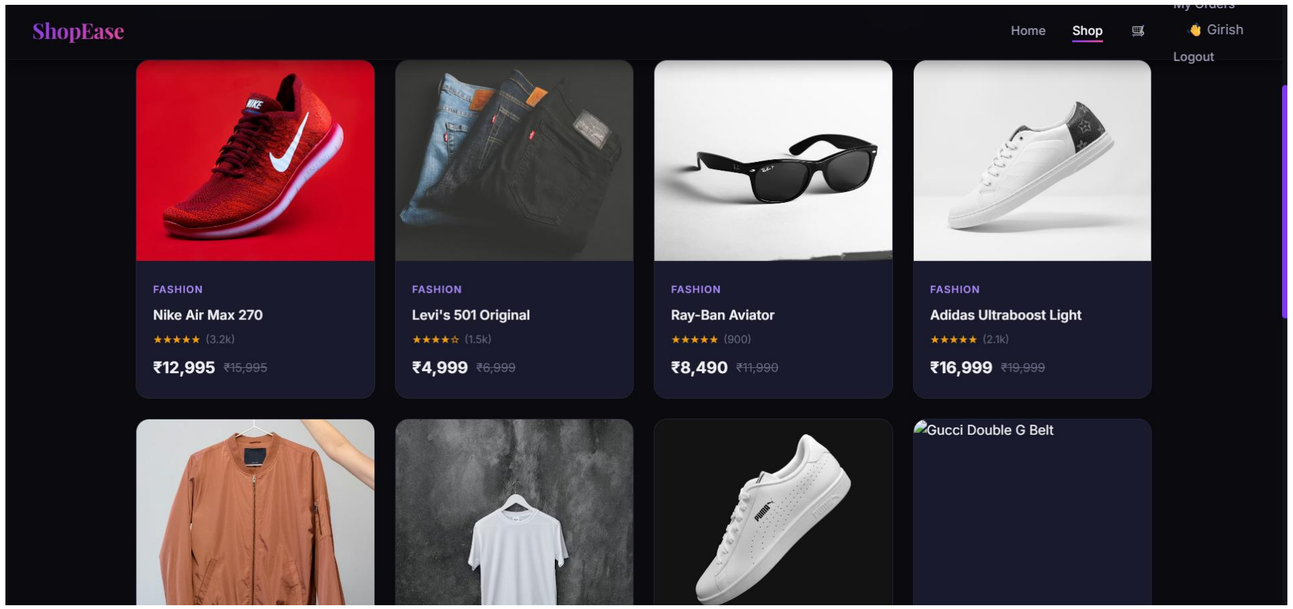
Module 4: Order Processing & Stock Management

- Places orders in a DB transaction: calculates total, saves order + items, deducts stock atomically, clears cart. Stores history per user via /api/orders/* endpoints.

IMPLEMENTATION SCREENSHOTS

- **Home Page:** Displays featured products (iPhone 15 Pro, MacBook Air M3) in card layout with image, name, discounted price, rating, and Add to Cart button.
- **Product Listing:** Shows all 12 products with category filter buttons. Products fetched from GET /api/products and rendered dynamically via JavaScript.
- **Search & Filter:** User types keyword and list updates in real-time via GET /api/products?search=keyword. Category buttons filter results instantly.
- **Product Card:** Each card shows: image, category badge, star rating, review count, current price (INR), original price strikethrough, discount badge, and action buttons.


ShopEase
Home
Shop
Girish
Logout
Our Products
Browse our curated collection
Search products...
Electronics
Price: High to Low
ELECTRONICS
Canon EOS R6 Mark II
(120)
₹2,29,900 ₹2,43,900
ELECTRONICS
Dell XPS 15
(320)
₹1,89,900 ₹2,10,000
ELECTRONICS
MacBook Air M3
(650)
₹1,49,900 ₹1,69,900
ELECTRONICS
iPhone 15 Pro Max
(1.2k)
₹1,34,900 ₹1,49,900


ShopEase
Home
Shop
Girish
Logout
FASHION
Nike Air Max 270
(3.2k)
₹12,995 ₹15,995
FASHION
Levi's 501 Original
(1.5k)
₹4,999 ₹6,999
FASHION
Ray-Ban Aviator
(900)
₹8,490 ₹11,990
FASHION
Adidas Ultraboost Light
(2.1k)
₹16,999 ₹19,999
Gucci Double G Belt

IMPLEMENTATION SCREENSHOTS

- **Login Page:** Form with Email and Password fields. POST /api/auth/login is called. Valid credentials create a session; invalid ones return an error message.
- **Registration Page:** Form with Name, Email, Password, Phone, Address. POST /api/auth/register is called. Duplicate email shows error; success redirects to login.
- **Session Management:** GET /api/auth/session is called on page load. If logged in, user name appears in navbar and cart badge shows item count.
- **Logout:** POST /api/auth/logout invalidates the server session. Frontend clears state and redirects to home, removing username from header.

Create Account

Join ShopEase for the best shopping experience

Full Name

John

Email Address

john@gmail.com

Password

••••••

Phone Number (Optional)

9078675423

Create Account

Already have an account? Login here

Welcome Back

Login to access your cart & orders

Email Address

john@gmail.com

Password

••••••

Login

Don't have an account? Register here

Demo: admin@shopease.com / admin123
Customer: manoj@example.com / customer123

IMPLEMENTATION SCREENSHOTS

- **Cart Page:** Lists all cart items from GET /api/cart. Each row shows product image, name, price, quantity stepper, subtotal, and Remove button. Total shown at bottom.
- **Add to Cart:** POST /api/cart with productId. Cart badge updates immediately. If item exists, quantity increments via ON DUPLICATE KEY UPDATE in MySQL.
- **Update Quantity:** PUT /api/cart with new quantity. If quantity hits 0, item is removed. Cart total recalculates instantly.
- **State Persistence:** Cart items stored in MySQL remain after browser close and re-login. This is the key feature differentiating ShopEase from session-based carts.

Shopping Cart

Your items are saved across sessions — powered by MySQL

Dyson V15 Detect **₹65,900**

₹65,900

− 1 +

MacBook Air M3 **₹1,49,900**

₹1,49,900

− 1 +

Select All

Order Summary

Selected Items (2)	₹2,15,800
Shipping	FREE

IMPLEMENTATION SCREENSHOTS

- **Checkout Page:** User enters shipping address and selects COD payment. POST /api/orders validates cart, places order, deducts stock atomically, and clears cart.
- **Order Confirmation:** Server returns orderId on success. Frontend shows Order placed successfully with Order ID.
- **Order History:** GET /api/orders returns all orders ordered by date descending. Each shows: order ID, status, total amount, payment method, and ordered items.
- **Order Item Details:** Each order expands to show product image, name, quantity, and price per item from order_items table joined with products via OrderDAO.

Logout

Shipping Details

Full Name

Girish

Phone Number

7675438977

Shipping Address

Avadi

Payment Method

Cash on Delivery (COD)

Place Order

Order Summary

Dyson V15 Detect ₹NaN

Qty: 1 × ₹65,900

Subtotal	₹65,900
Shipping	FREE
Total	**₹65,900**

ShopEase

Premium E-Commerce Experience

INVOICE

#ORD-15

Date: 3/4/2026

CUSTOMER DETAILS

Girish
girish@gmail.com
7673486521

SHIPPING ADDRESS

Delhi

EXPECTED DELIVERY

3/8/2026

PRODUCT	PRICE	QTY	TOTAL
Adidas Ultraboost Light	₹16,999	1	₹16,999

Subtotal:	₹16,999
Tax (5% GST):	₹849.95
Shipping Charges:	₹0

Total Payable: ₹17,848.95

ADVANTAGES & LIMITATIONS

ADVANTAGES:

- State-persistent cart stored in MySQL — items remain even after browser close, logout, or device change.
- BCrypt password hashing ensures user passwords are never stored in plain text, providing strong security.
- Embedded Tomcat via Spring Boot means no separate server installation or configuration is required.
- RESTful API design (AuthServlet, ProductServlet, CartServlet, OrderServlet) makes the backend modular and extensible.
- Atomic database transactions for order placement prevent overselling and ensure data consistency.
- Role-based access control separates CUSTOMER and ADMIN privileges cleanly within the same application.

LIMITATIONS:

- No integrated payment gateway — currently only supports Cash on Delivery (COD) as payment method.
- Product images are referenced via external URLs; no image upload feature is available in this milestone.
- Admin panel features are limited; a full admin dashboard with analytics is planned for a future milestone.

Thank you